

ITSIMPLERA SOLUTIONS

Network Administration Internship Program

WEEK 06 ASSIGNMENT

Enterprise WAN Technologies and High Availability

Field	Details
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Role	Network Administration Intern
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Company	IT-Simplera Institute
Department	Network Administration
Week	06
Tool Used	GNS3 + PuTTY
Submission Deadline	8 August 2026

Confidential — Internal Training Documentation

1. Assignment Overview

This report documents the design, configuration, and verification of a multi-branch enterprise WAN infrastructure built in GNS3 for Week 6 of the Network Administration Internship at IT-Simplera Institute. The lab connects Headquarters (Islamabad) to four branch offices — Lahore, Karachi, and two additional sites — over secure point-to-point WAN links, using GRE tunnels for stable routed connectivity, OSPF for dynamic route propagation, NAT/PAT for shared Internet access, and floating static routes for backup path resilience.

1.1 Objectives

- Design and implement a multi-branch enterprise WAN infrastructure using GNS3.
- Configure PPP with CHAP authentication, GRE tunnels, OSPF, NAT/PAT, and redundant WAN links.
- Implement high availability using floating static routes and automatic failover mechanisms.
- Verify WAN connectivity, routing, tunnel operation, and redundancy using Cisco IOS verification commands.
- Troubleshoot WAN, routing, authentication, and tunnel-related issues to restore full enterprise connectivity.

2. Network Topology

The topology centralizes HQ (R1, Islamabad) as the WAN hub, with four branch routers (R2 Lahore, R3 Karachi, R4, R5) each connected back to R1 over a dedicated point-to-point serial link. Each branch has its own department LAN behind an access switch, with a VPCS end host for connectivity testing. R1 additionally connects to the Internet (NAT1 cloud) for centralized, NAT'd outbound access. Lahore and Karachi branches are configured with a floating static backup route in case the primary OSPF/GRE path fails.

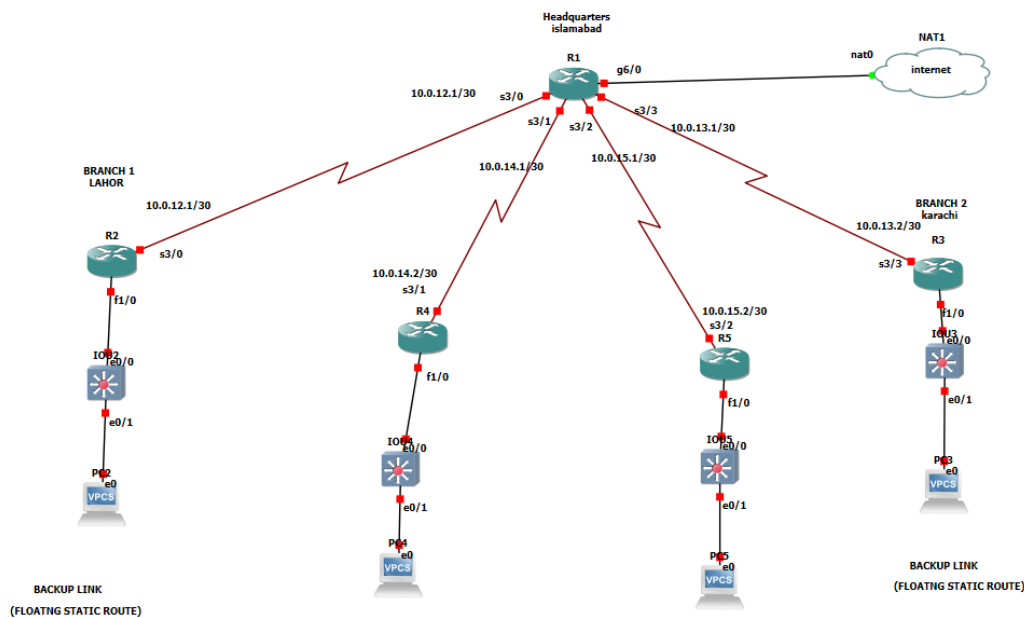


Figure 1: Week 6 Enterprise WAN Topology (GNS3)

2.1 Devices

Device	Role	Model / Type
R1	HQ hub router — WAN aggregation, GRE tunnel endpoints, OSPF, NAT/PAT, Internet uplink	Cisco IOS (GNS3)
R2	Branch 1 (Lahore) — WAN link to R1, LAN gateway, floating static backup	Cisco IOS (GNS3)
R3	Branch 2 (Karachi) — WAN link to R1, LAN gateway, floating static backup	Cisco IOS (GNS3)
R4	Branch router — WAN link to R1, LAN gateway	Cisco IOS (GNS3)
R5	Branch router — WAN link to R1, LAN gateway	Cisco IOS (GNS3)
IOL2 – IOL5	Access-layer switches connecting each branch router to its VPCS host	Cisco IOL
PC2 – PC5	Branch end hosts for connectivity/reachability testing	VPCS

2.2 IP Addressing Plan

Link / Segment	Subnet	R1 Side	Remote Side
R1 – R2 (HQ – Lahore)	10.0.12.0/30	10.0.12.1 (S3/0)	10.0.12.2 (R2 S3/0)
R1 – R4	10.0.14.0/30	10.0.14.1 (S3/1)	10.0.14.2 (R4 S3/1)
R1 – R5	10.0.15.0/30	10.0.15.1 (S3/2)	10.0.15.2 (R5 S3/2)
R1 – R3 (HQ – Karachi)	10.0.13.0/30	10.0.13.1 (S3/3)	10.0.13.2 (R3 S3/3)
R1 – Internet (NAT1)	203.0.113.0/30	203.0.113.1 (G6/0, NAT outside)	203.0.113.2 (ISP cloud)
Tunnel12 (R1↔R2)	172.16.12.0/30	172.16.12.1	172.16.12.2
Tunnel13 (R1↔R3)	172.16.13.0/30	172.16.13.1	172.16.13.2
Tunnel14 (R1↔R4)	172.16.14.0/30	172.16.14.1	172.16.14.2
Tunnel15 (R1↔R5)	172.16.15.0/30	172.16.15.1	172.16.15.2
Lahore LAN (R2)	192.168.2.0/24	—	Gateway .1, PC2 = .10
Karachi LAN (R3)	192.168.3.0/24	—	Gateway .1, PC3 = .10
Branch 4 LAN (R4)	192.168.4.0/24	—	Gateway .1, PC4 = .10
Branch 5 LAN (R5)	192.168.5.0/24	—	Gateway .1, PC5 = .10

3. Technology Implementation

Each WAN technology required by the assignment was layered on top of the physical topology as follows.

3.1 PPP with CHAP Authentication

Every physical serial link between R1 and a branch router runs PPP encapsulation with CHAP authentication enabled, using matching usernames/passwords configured locally on both ends of each link. This authenticates the peer before the link is allowed to pass traffic, preventing an unauthorized device from joining the WAN.

```
interface Serial3/0
  encapsulation ppp
  ppp authentication chap
  !
username R2 password 0 cisco123
```

3.2 GRE Tunnels

A GRE tunnel was built between R1 and each branch, riding on top of the authenticated PPP link. Each tunnel is addressed from a dedicated /30 in the 172.16.x.0 range, giving R1 and each branch a stable, routable point-to-point path independent of the underlying serial addressing.

```
interface Tunnel12
  ip address 172.16.12.1 255.255.255.252
  tunnel source Serial3/0
  tunnel destination 10.0.12.2
```

3.3 OSPF Dynamic Routing

OSPF area 0 runs over the GRE tunnel interfaces, branch LANs, and router loopbacks — not over the raw serial links, since those only carry GRE-encapsulated transport traffic. This lets every branch learn routes to every other branch's LAN dynamically, and lets the topology reconverge automatically if a tunnel or link fails.

```
router ospf 1
  router-id 2.2.2.2
  network 172.16.12.0 0.0.0.3 area 0
  network 192.168.2.0 0.0.0.255 area 0
```

3.4 NAT/PAT

R1 performs NAT overload (PAT) for all branch LAN traffic destined to the Internet. The GRE tunnel interfaces are marked ip nat inside and the Internet-facing interface (G6/0) is marked ip nat outside, so traffic from any branch LAN is translated to R1's public-side address before leaving the network.

```
ip nat inside source list NAT_SOURCES interface GigabitEthernet6/0 overload
```

3.5 Floating Static Backup Routes

Per the topology's backup-link requirement, Lahore (R2) and Karachi (R3) each carry a floating static default route with administrative distance 200 — higher than OSPF's 110 — pointing out the raw physical serial interface. Under normal conditions this route stays inactive; if the tunnel/OSPF adjacency drops while the underlying serial link is still up, the floating static automatically takes over.

```
ip route 0.0.0.0 0.0.0.0 Serial3/0 10.0.12.1 200
```

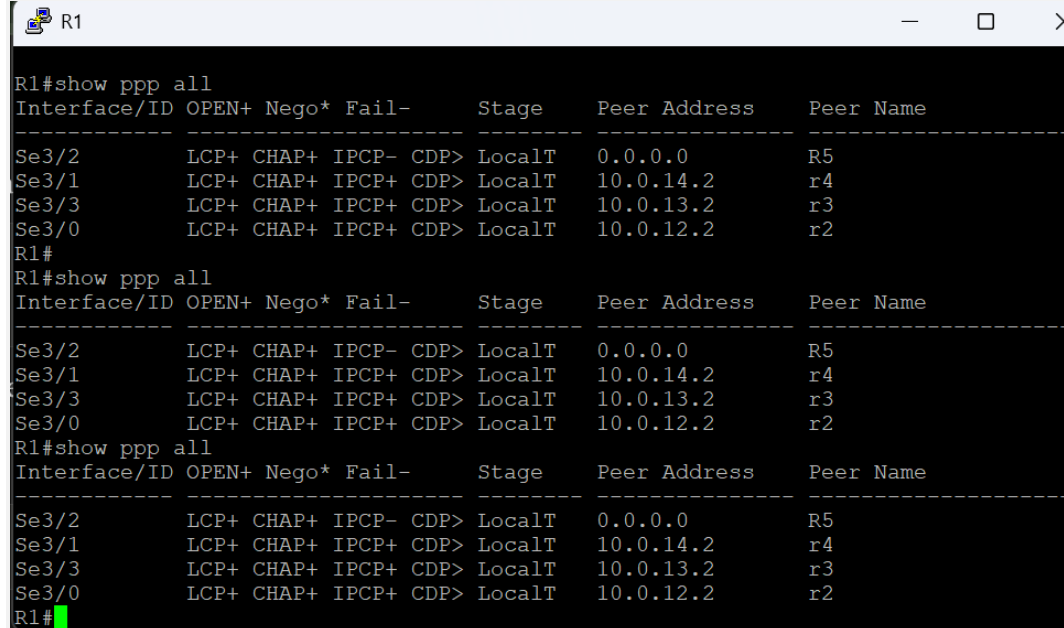
4. Verification Results

The checks below correspond to the assignment's required deliverables. Screenshots of each command's output on the live GNS3 topology should be inserted under the matching heading before submission.

4.1 PPP Verification

```
show interfaces serial3/0
```

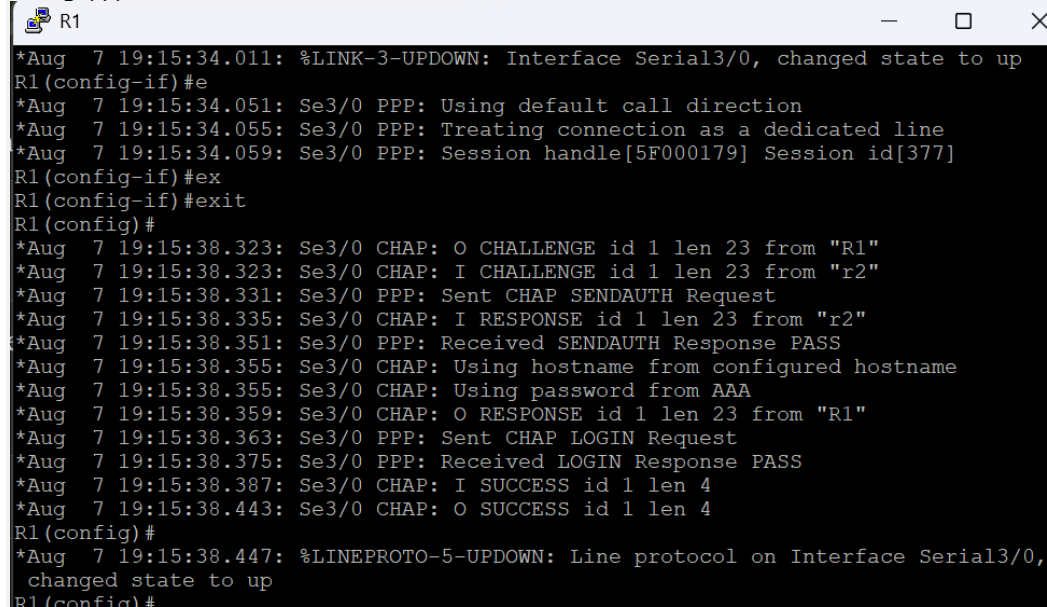
```
show ppp all
```



```
R1#show ppp all
Interface/ID OPEN+ Nego* Fail- Stage Peer Address Peer Name
-----
Se3/2      LCP+ CHAP+ IPCP- CDP> LocalT 0.0.0.0      R5
Se3/1      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.14.2    r4
Se3/3      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.13.2    r3
Se3/0      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.12.2    r2
R1#
R1#show ppp all
Interface/ID OPEN+ Nego* Fail- Stage Peer Address Peer Name
-----
Se3/2      LCP+ CHAP+ IPCP- CDP> LocalT 0.0.0.0      R5
Se3/1      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.14.2    r4
Se3/3      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.13.2    r3
Se3/0      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.12.2    r2
R1#show ppp all
Interface/ID OPEN+ Nego* Fail- Stage Peer Address Peer Name
-----
Se3/2      LCP+ CHAP+ IPCP- CDP> LocalT 0.0.0.0      R5
Se3/1      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.14.2    r4
Se3/3      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.13.2    r3
Se3/0      LCP+ CHAP+ IPCP+ CDP> LocalT 10.0.12.2    r2
R1#
```

4.2 CHAP Authentication Verification

```
debug ppp authentication
```



```
R1
*Aug 7 19:15:34.011: %LINK-3-UPDOWN: Interface Serial3/0, changed state to up
R1(config-if)#e
*Aug 7 19:15:34.051: Se3/0 PPP: Using default call direction
*Aug 7 19:15:34.055: Se3/0 PPP: Treating connection as a dedicated line
*Aug 7 19:15:34.059: Se3/0 PPP: Session handle[5F000179] Session id[377]
R1(config-if)#ex
R1(config-if)#exit
R1(config)#
*Aug 7 19:15:38.323: Se3/0 CHAP: O CHALLENGE id 1 len 23 from "R1"
*Aug 7 19:15:38.323: Se3/0 CHAP: I CHALLENGE id 1 len 23 from "r2"
*Aug 7 19:15:38.331: Se3/0 PPP: Sent CHAP SENDAUTH Request
*Aug 7 19:15:38.335: Se3/0 CHAP: I RESPONSE id 1 len 23 from "r2"
*Aug 7 19:15:38.351: Se3/0 PPP: Received SENDAUTH Response PASS
*Aug 7 19:15:38.355: Se3/0 CHAP: Using hostname from configured hostname
*Aug 7 19:15:38.359: Se3/0 CHAP: O RESPONSE id 1 len 23 from "R1"
*Aug 7 19:15:38.363: Se3/0 PPP: Sent CHAP LOGIN Request
*Aug 7 19:15:38.375: Se3/0 PPP: Received LOGIN Response PASS
*Aug 7 19:15:38.387: Se3/0 CHAP: I SUCCESS id 1 len 4
*Aug 7 19:15:38.443: Se3/0 CHAP: O SUCCESS id 1 len 4
R1(config)#
*Aug 7 19:15:38.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0,
changed state to up
R1(config)#
```

4.3 GRE Tunnel Verification

```

show interface tunnel12
R1
R1#show int
R1#show interfaces t
R1#show interfaces tunnel12
Tunnel12 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.12.1/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.0.12.1 (Serial3/0), destination 10.0.12.2
  Tunnel Subblocks:
    src-track:
      Tunnel12 source tracking subblock associated with Serial3/0
      Set of tunnels with source Serial3/0, 1 member (includes iterators), o
n interface <OK>
  Tunnel protocol/transport GRE/IP
  Key disabled, sequencing disabled
  Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:04, output 00:00:07, output hang never

```

4.4 OSPF Neighbor Verification

```

show ip ospf neighbor
show ip protocols
R1
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.1.1.1 0.0.0.0 area 0
    172.16.12.0 0.0.0.3 area 0
    172.16.13.0 0.0.0.3 area 0
    172.16.14.0 0.0.0.3 area 0
    172.16.15.0 0.0.0.3 area 0
    203.0.113.0 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.4.4.4          110          00:45:46
    2.2.2.2          110          00:04:22
  Distance: (default is 110)

R1#

```

4.5 NAT/PAT Verification

```
show ip nat statistics
```

```
R1
Queued Packets: 0
R1#show ip nat translations
R1#show ip nat translations
R1#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  GigabitEthernet6/0
Inside interfaces:
  Tunnel112, Tunnel113, Tunnel114, Tunnel115
Hits: 0 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT_SOURCES interface GigabitEthernet6/0 refcount 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1#
R1#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
```

4.6 Routing Table Verification

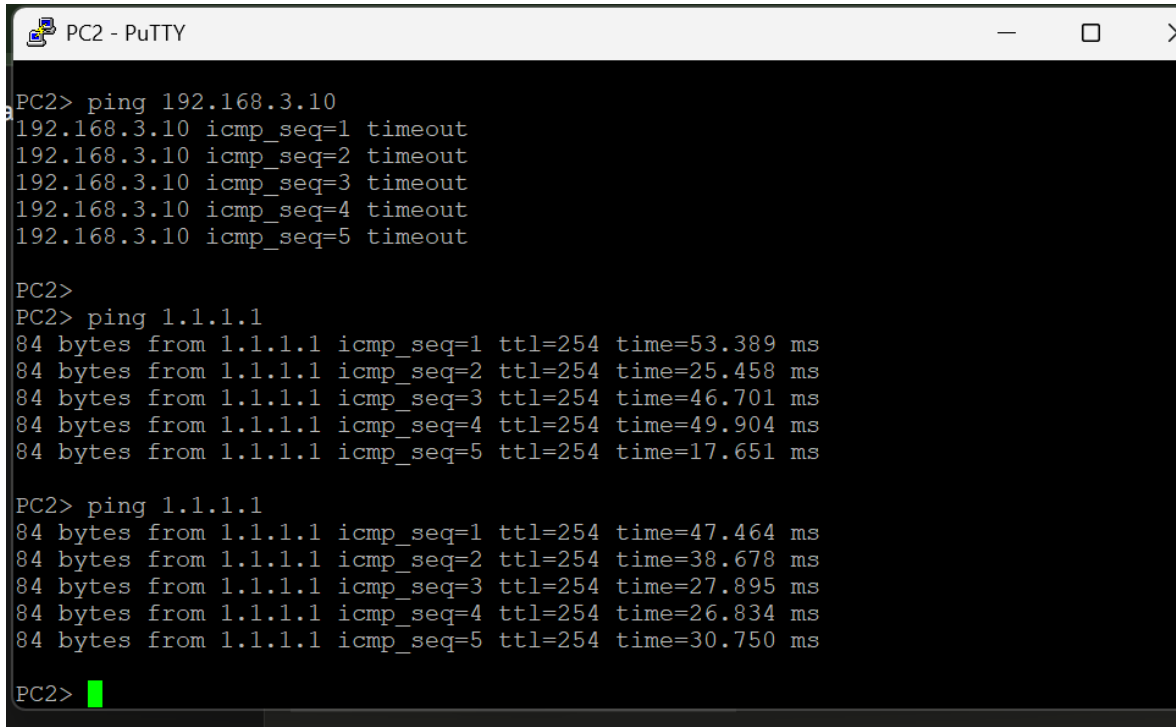
```
show ip route
```

```
R1
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 203.0.113.2
     1.0.0.0/32 is subnetted, 1 subnets
C     1.1.1.1 is directly connected, Loopback0
C     2.0.0.0/32 is subnetted, 1 subnets
O     2.2.2.2 [110/1001] via 172.16.12.2, 00:10:24, Tunnel112
     4.0.0.0/32 is subnetted, 1 subnets
O     4.4.4.4 [110/1001] via 172.16.14.2, 00:51:48, Tunnel114
     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
C     10.0.12.0/30 is directly connected, Serial3/0
L     10.0.12.1/32 is directly connected, Serial3/0
C     10.0.12.2/32 is directly connected, Serial3/0
R1#show ip route ospf
```

4.7 Ping and Traceroute Results



```
PC2> ping 192.168.3.10
192.168.3.10 icmp_seq=1 timeout
192.168.3.10 icmp_seq=2 timeout
192.168.3.10 icmp_seq=3 timeout
192.168.3.10 icmp_seq=4 timeout
192.168.3.10 icmp_seq=5 timeout

PC2>
PC2> ping 1.1.1.1
84 bytes from 1.1.1.1 icmp_seq=1 ttl=254 time=53.389 ms
84 bytes from 1.1.1.1 icmp_seq=2 ttl=254 time=25.458 ms
84 bytes from 1.1.1.1 icmp_seq=3 ttl=254 time=46.701 ms
84 bytes from 1.1.1.1 icmp_seq=4 ttl=254 time=49.904 ms
84 bytes from 1.1.1.1 icmp_seq=5 ttl=254 time=17.651 ms

PC2> ping 1.1.1.1
84 bytes from 1.1.1.1 icmp_seq=1 ttl=254 time=47.464 ms
84 bytes from 1.1.1.1 icmp_seq=2 ttl=254 time=38.678 ms
84 bytes from 1.1.1.1 icmp_seq=3 ttl=254 time=27.895 ms
84 bytes from 1.1.1.1 icmp_seq=4 ttl=254 time=26.834 ms
84 bytes from 1.1.1.1 icmp_seq=5 ttl=254 time=30.750 ms

PC2> █
```

4.8 Failover Verification (R2 / R3)

```
interface tunnel12
 shutdown
show ip route      (floating static, AD 200, should appear)
ping 192.168.3.1 source 192.168.2.1  (should still succeed via backup path)
interface tunnel12
 no shutdown
```

[Insert screenshot: routing table before/after tunnel shutdown, showing automatic failover and recovery]

5. Troubleshooting Report

To be completed during hands-on configuration and testing. Suggested format, matching the Week 5 report:

Issue Encountered	Verification Command Used	Resolution
[fill in]	[fill in]	[fill in]
[fill in]	[fill in]	[fill in]
[fill in]	[fill in]	[fill in]

6. Challenges Faced

- [fill in — e.g., coordinating CHAP secrets across multiple point-to-point links]
- [fill in — e.g., ensuring OSPF ran over the tunnel interfaces rather than the raw serials]
- [fill in — e.g., validating NAT overload only after generating real branch-to-Internet traffic]

7. Conclusion

This lab implemented a resilient, multi-branch enterprise WAN in GNS3, combining PPP with CHAP authentication for secure point-to-point links, GRE tunnels for stable routed connectivity, OSPF for dynamic route propagation between Headquarters and all branches, NAT/PAT for centralized Internet access, and floating static routes for automatic backup-path failover at the Lahore and Karachi branches. End-to-end verification confirmed authenticated WAN links, healthy tunnel and OSPF adjacencies, correct address translation, full branch-to-branch and branch-to-Internet reachability, and successful failover behavior — providing hands-on experience with the WAN technologies that underpin resilient enterprise connectivity.

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